

## Self-test Exercise

1. Which of the following is true of standard error?
  - a. It can take negative values.
  - b. It is an estimate of the standard deviation.
  - c. It is the square root of the variance.
  - d. It complicates the computation of confidence intervals.

2. In the following equation, gdp refers to gross domestic product, and FDI refers to foreign direct investment.

$$\widehat{\log(\text{gdp})} = 2.65 + 0.527\log(\text{bankcredit}) + 0.222\text{FDI}$$

(0.13)   (0.022)                      (0.017)

Which of the following statements is then true?

- a. If gdp increases by 1%, bank credit increases by 0.527%, the level of FDI remaining constant.
- b. If bank credit increases by 1%, gdp increases by 0.527%, the level of FDI remaining constant.
- c. If gdp increases by 1%, bank credit increases by  $\log(0.527)\%$ , the level of FDI remaining constant.
- d. If bank credit increases by 1%, gdp increases by  $\log(0.527)\%$ , the level of FDI remaining constant.

3. Which of the following statements is true when the dependent variable,  $y > 0$ ?

- a. Taking log of a variable often expands its range.
- b. Models using  $\log(y)$  as the dependent variable will satisfy CLM assumptions more closely than models using the level of  $y$ .
- c. Taking log of variables make OLS estimates more sensitive to extreme values.
- d. Taking logarithmic form of variables make the slope coefficients more responsive to rescaling.

4. Which of the following models is used quite often to capture decreasing or increasing marginal effects of a variable?

- a. Models with logarithmic functions
- b. Models with quadratic functions
- c. Models with variables in level
- d. Models with interaction terms

5. A predicted value of a dependent variable:

- a. represents the difference between the expected value of the dependent variable and its actual value.
- b. is always equal to the actual value of the dependent variable.
- c. is independent of explanatory variables and can be estimated on the basis of the residual error term only.
- d. represents the expected value of the dependent variable given particular values for the explanatory variables.

6. If the R-squared value is low, then using OLS equation is very easy to predict individual future outcomes on y given a set of values for the explanatory variables.

- a. True
- b. False

7. In a regression model, which of the following will be described using a binary variable?

- a. Whether it rained on a particular day or it did not
- b. The volume of rainfall during a year
- c. The percentage of humidity in air on a particular day
- d. The concentration of dust particles in air

8. The following simple model is used to determine the annual savings of an individual on the basis of his annual income and education.

$\text{Savings} = \beta_0 + \beta_1 \text{Edu} + \beta_2 \text{Inc} + u$ . The variable 'Edu' takes a value of 1 if the person is educated and the variable 'Inc' measures the income of the individual.

Refer to the model above. The inclusion of another binary variable in this model that takes a value of 1 if a person is uneducated, will give rise to the problem of \_\_\_\_\_.

- a. omitted variable bias
- b. self-selection
- c. dummy variable trap
- d. heteroskedasticity

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Refer to the model above. The benchmark group (or the base group) in this model is \_\_\_\_\_.

- a. the group of educated people
- b. the group of uneducated people
- c. the group of individuals with a high income
- d. the group of individuals with a low income

10. The income of an individual in Country A depends on his ethnicity and several other factors which can be measured quantitatively.

If there are 5 ethnic groups in Country A, how many dummy variables should be included in the regression equation with an intercept for income determination in Country A?

- a. 1
- b. 5
- c. 6
- d. 4

11. Which of the following is true of dependent variables?
- A dependent variable can only have a numerical value.
  - A dependent variable cannot have more than 2 values.
  - A dependent variable can be binary.
  - A dependent variable cannot have a qualitative meaning.
12. If the p-value of an F statistic 2.63 is 0.034, then we can say that the problem of interest is significant at the 5% level.
- True
  - False
13. The multiple linear regression model with a binary dependent variable is called the linear probability model.
- True
  - False
14. Which of the following is true of heteroskedasticity?
- Heteroskedasticity causes inconsistency in the Ordinary Least Squares estimators.
  - Population  $R^2$  is affected by the presence of heteroskedasticity.
  - The Ordinary Least Square estimators are not the best linear unbiased estimators if heteroskedasticity is present.
  - It is not possible to obtain F statistics that are robust to heteroskedasticity of an unknown form.
15. Which of the following tests helps in the detection of heteroskedasticity?
- The Breusch-Pagan test
  - The Breusch-Godfrey test
  - The Durbin-Watson test
  - The Chow test
16. What will you conclude about a regression model if the Breusch-Pagan test results in a small p-value?
- The model contains homoskedasticity.
  - The model contains heteroskedasticity.
  - The model contains dummy variables.
  - The model omits some important explanatory factors.